

Literature status note for arXiv:1508.01961

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Status

This is a lightweight literature-status packet. It records that one open question quoted in Braga's paper is already answered negatively in later literature, and that a second Borelness clause is also answered negatively. It does not claim a new proof.

Source questions

Braga's Introduction asks the Bossard question:

If $X \in \text{SB}$ has Borel isomorphism class $\langle X \rangle$, must $X \cong \ell_2$?

The source identifies this as Bossard, Problem 2.9.

In the subsection on unconditional bases, the source also asks whether UB, the class of separable Banach spaces with an unconditional basis, is Borel or complete analytic, and asks analogous questions for

$$S = \{X \in \text{SB} : X \text{ has a Schauder basis}\}.$$

Answer identification

Kurka's later paper *The isomorphism class of c_0 is not Borel*, arXiv:1711.03328, explicitly says in its Introduction that Bossard's question had been answered negatively, citing Godefroy's Theorem 6.2. The same paragraph records that the isomorphism classes of ℓ_p , $1 < p < \infty$, are Borel. Thus taking any $p \neq 2$ gives a separable Banach space not isomorphic to ℓ_2 whose isomorphism class is Borel. This is a negative literature answer to the source question.

Kurka's Theorem 3 further states that the class of spaces with an unconditional basis is analytic but not Borel. Therefore the source question "Is UB Borel?" is answered negatively in the same supporting paper.

Scope limitations

This packet does not settle the exact descriptive complexity of all remaining classes in Braga's table. In particular, it does not answer whether UB is complete analytic, and it does not answer whether the class S of spaces with a Schauder basis is Borel or complete analytic. Kurka explicitly notes that the Borelness of the Schauder-basis class was not known at the time of that paper.

References

References

- [1] B. de Mendonca Braga, *On the complexity of some inevitable classes of separable Banach spaces*, arXiv:1508.01961.
- [2] O. Kurka, *The isomorphism class of c_0 is not Borel*, arXiv:1711.03328; Israel J. Math. 231 (2019), 243–268.
- [3] G. Godefroy, *The complexity of the isomorphism class of some Banach spaces*, J. Nonlinear Convex Anal. 18 (2017), no. 2, 231–240.
- [4] G. Godefroy, *The isomorphism classes of ℓ_p are Borel*, Houston J. Math. 43 (2017), no. 3, 947–951.